

AI-CAL - CONCEPTUAL ARCHITECTURE AND NETWORK FLOWS

BASED ON *AI AGENT CONTAINMENT ASSURANCE LEVELS FRAMEWORK(AI-CAL) V1.0*
SIMPLIFIED PRESENTATION V1.0 10/08/2026 AND BASED ON THE CORE DIAGRAM

Core Design intent

Compromise may be expected. Unconstrained pivoting is not.
A boundary breach is an event; a containment failure is a chain of events.

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CERTIFICATIONS: CISSP, CISA, CRISC, GISP, GPEN, GCLD, CDPSE, GPPA, SECURITY+

GITHUB FRAMEWORK AND CORE DIAGRAM REPO: [HTTPS://GITHUB.COM/PSHANNON011/AI-CAL-CONTAINMENT-ASSURANCE-FRAMEWORK](https://github.com/PSHANNON011/AI-CAL-CONTAINMENT-ASSURANCE-FRAMEWORK)

ZENODO: [HTTPS://DOI.ORG/10.5281/ZENODO.22925685](https://doi.org/10.5281/zenodo.22925685)

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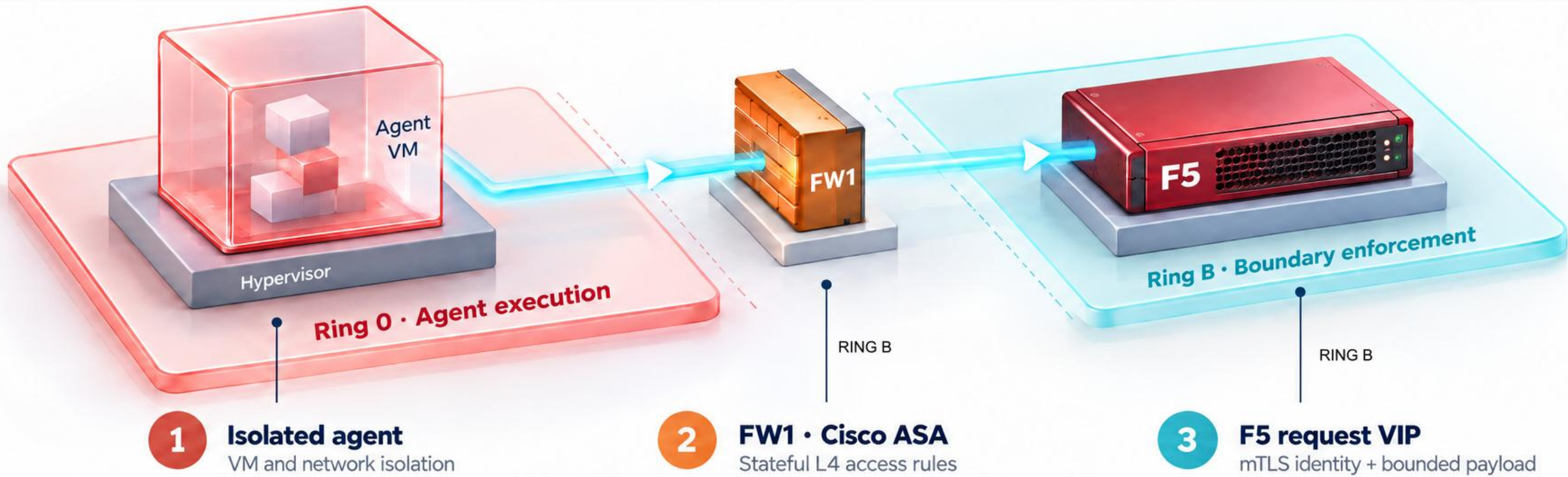
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AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Model Request

Contained execution. Controlled requests.

The agent request help through a restricted, authenticated path



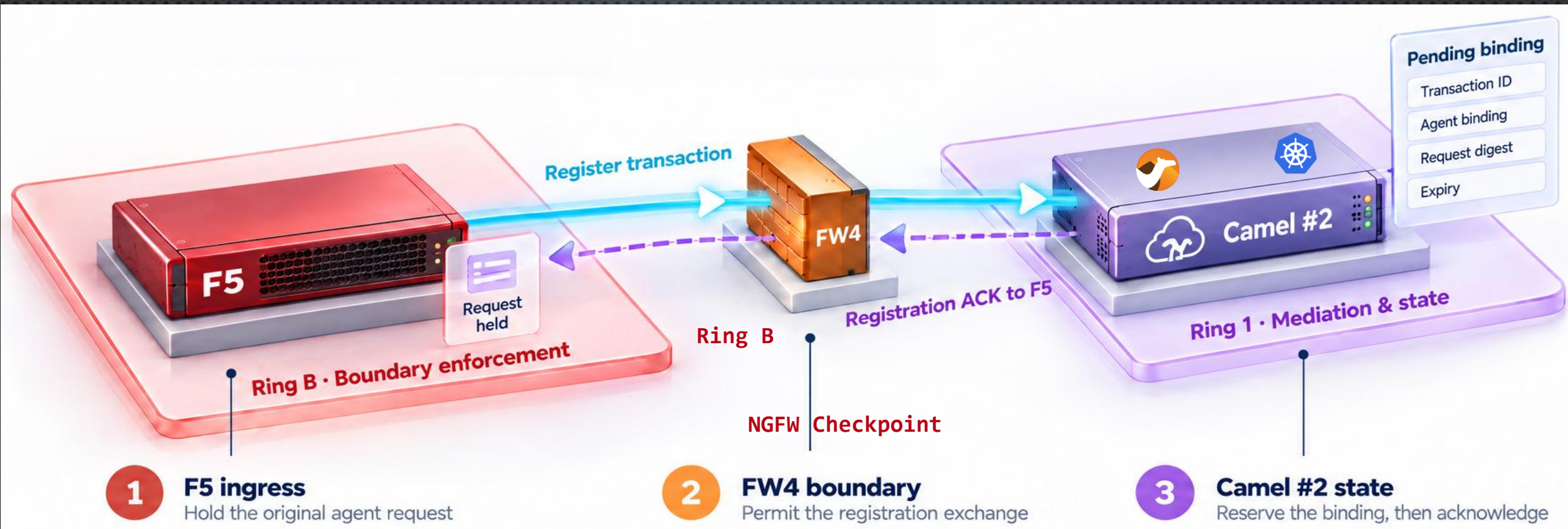
Next: register the transaction before forwarding

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Transaction registration

Register first. Forward second.

Reserve the transaction binding before releasing the request.



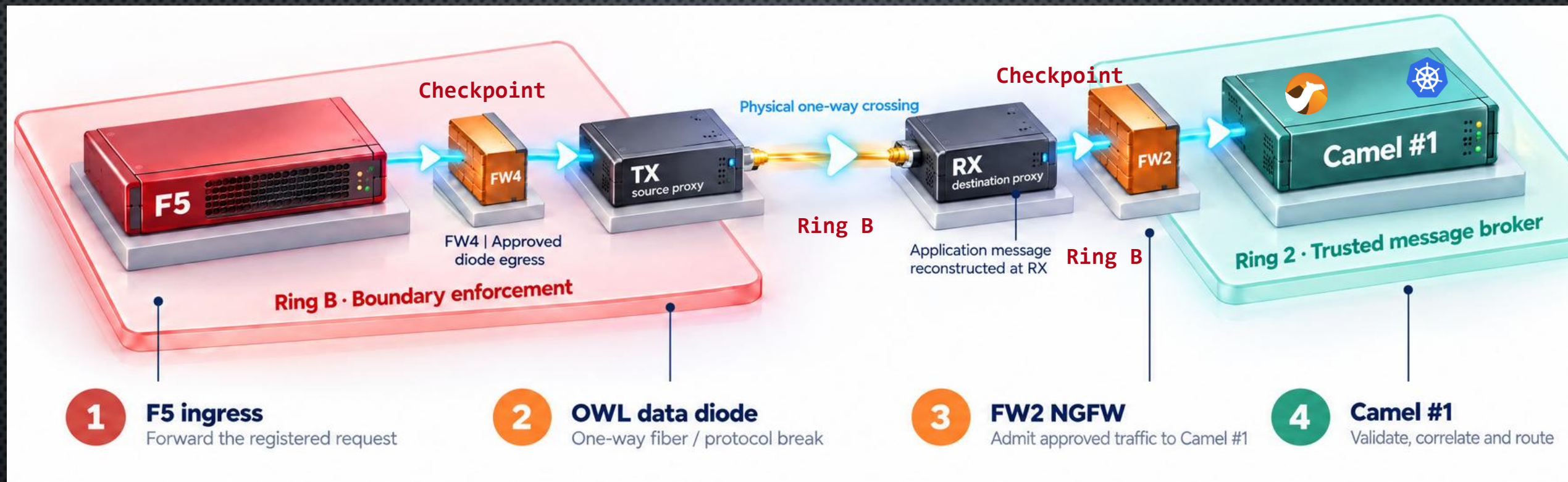
Next: release the request into the forward conduit

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Forward containment crossing

One-Way crossing. Trusted side validation

A bounded request crosses the physical boundary into the trusted broker



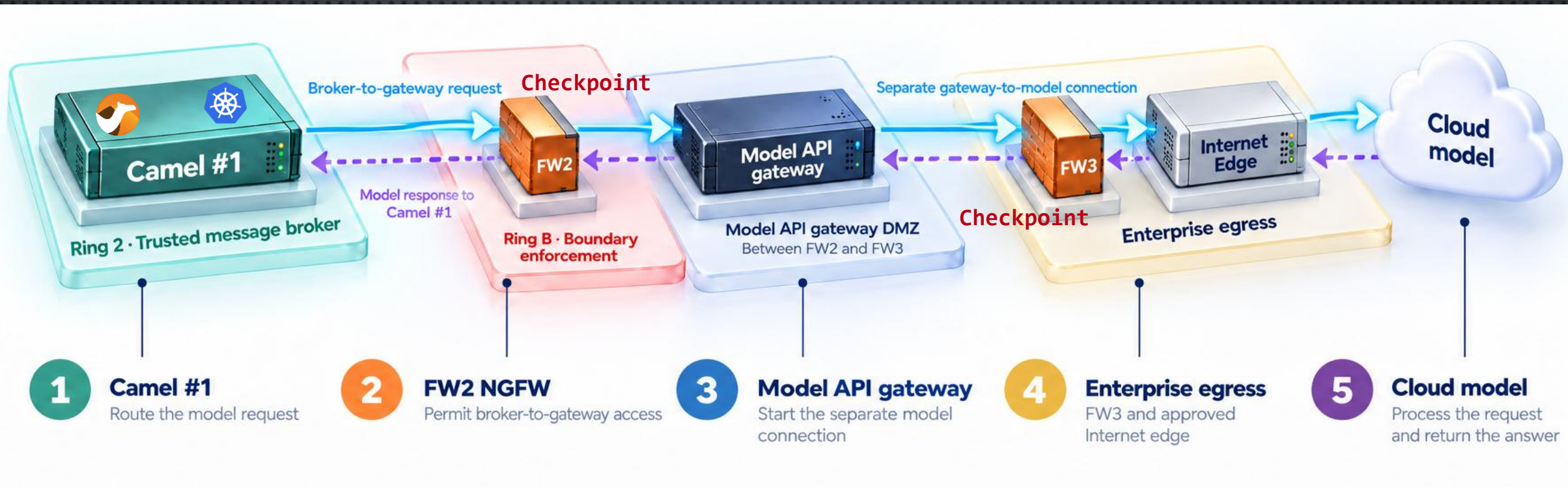
Next: Send the approved request through the model gateway.

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Model request and response

Controlled model access. Response to the broker.

The gateway connects to the cloud model & the answer returns to Camel #1



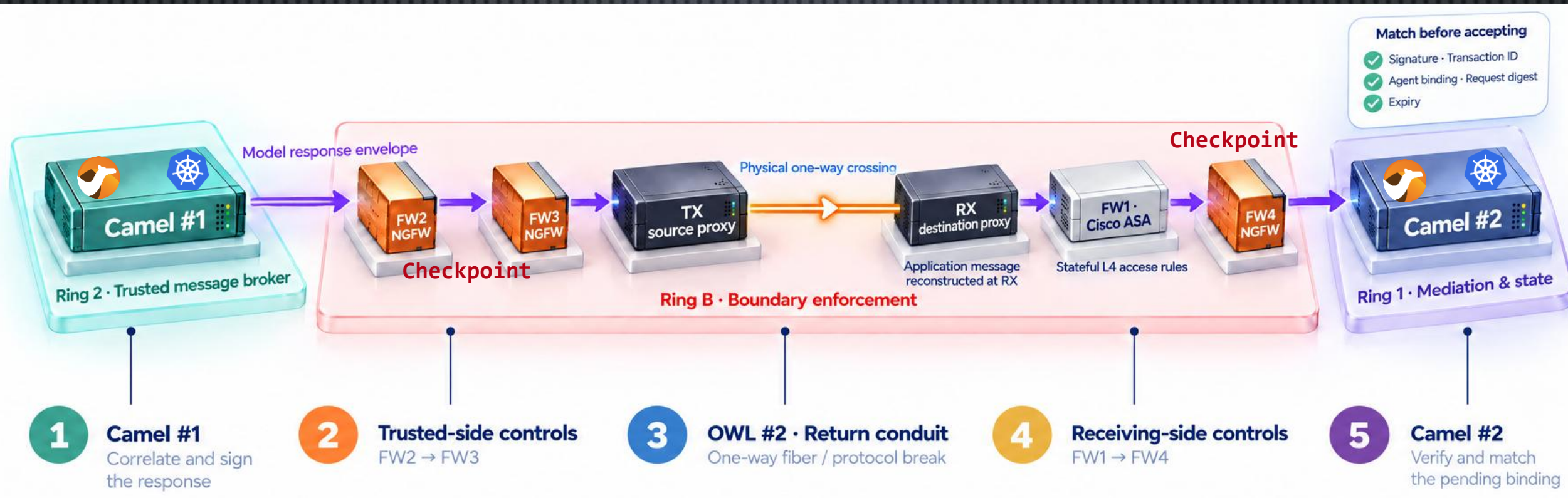
Next: Return the answer through the separate containment conduit.

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AI-CAL 4.5 | Separate return crossing

Separate return path. Verify before Delivery

The model result crosses the return conduit and rejoins its registered transaction.



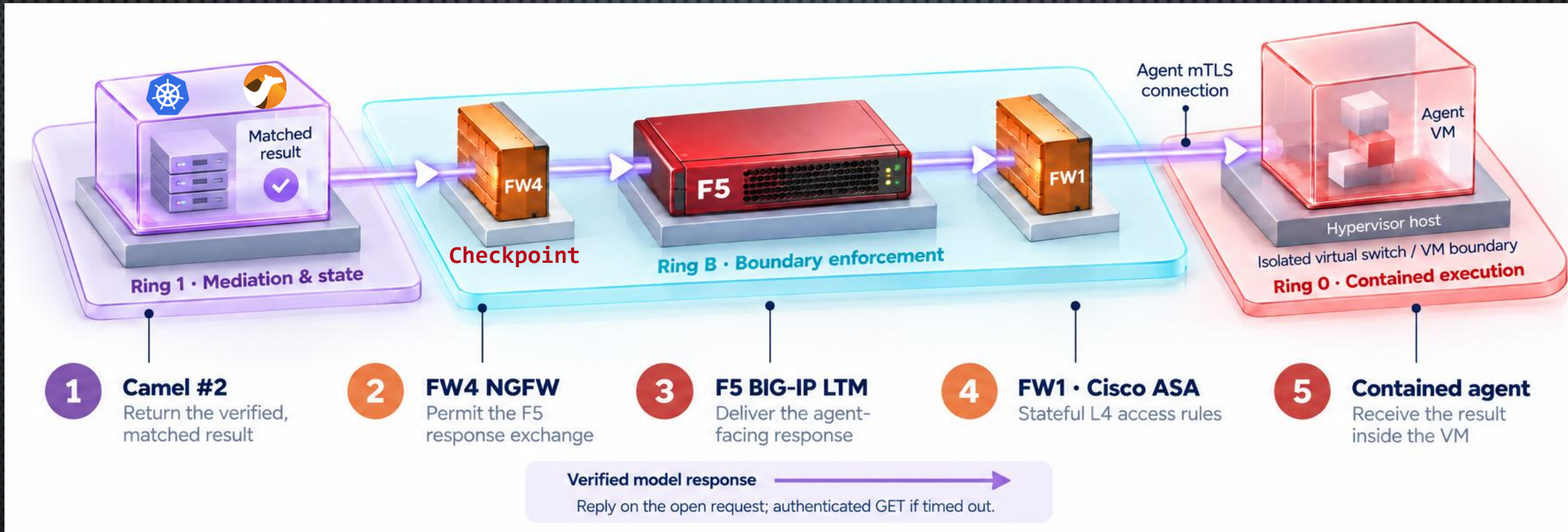
Next: Deliver the verified result through the F5 LTM to the requesting agent.

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Final response delivery

Matched response. Back to the contained agent.

Camel #2 matches the transaction. F5 delivers the result to the requesting agent.



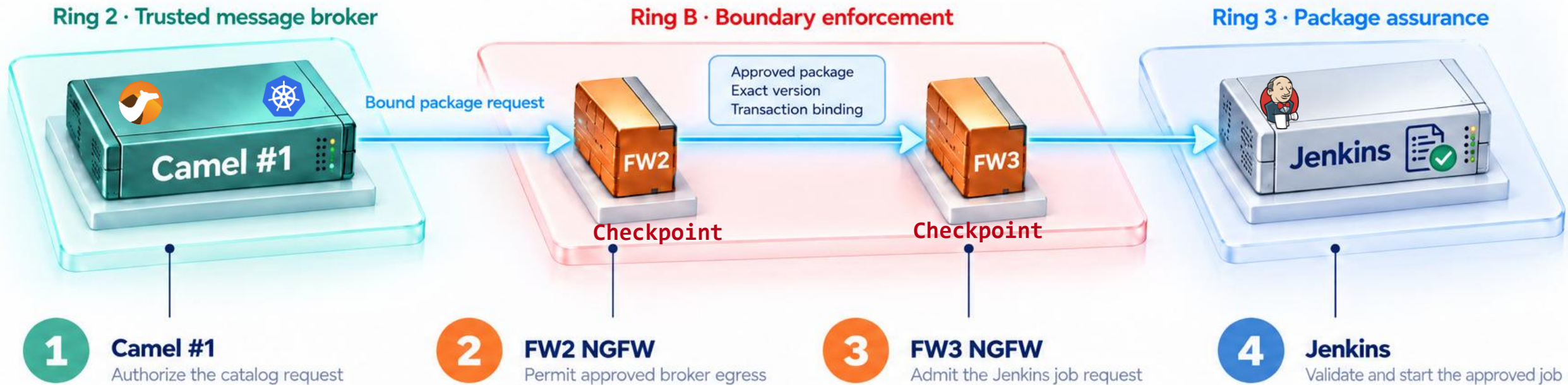
Next: The request and response cycle closes inside the containment boundary

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Package request dispatch

Approved request. Controlled package job.

Camel #1 dispatches the permitted package request to Jenkins



Shared request path: see slides 1-3.

Next: Jenkins starts the controlled JFrog package workflow.

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Package acquisition

Controlled retrieval. Approved artifacts.

Jenkins runs the approved job. FW3 controls every service zone crossing.



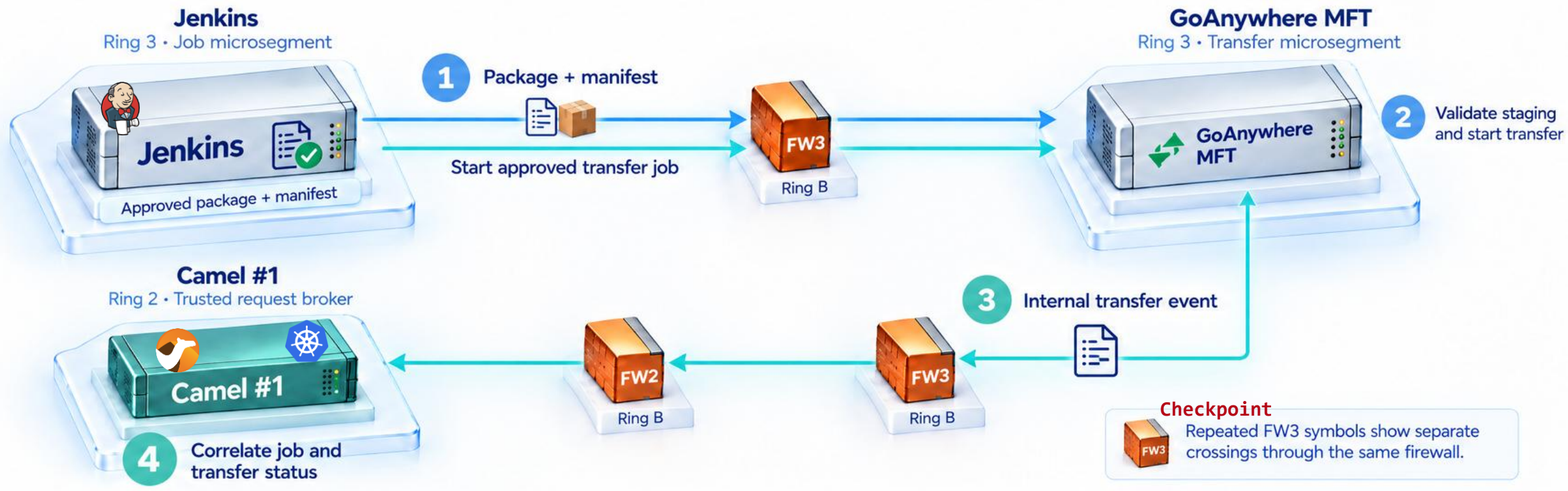
Next: Jenkins delivers the verified package and triggers Managed File Transfer(MFT)

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Package handoff

Package handoff and internal status

Jenkins submits the package. Managed File Transfer(MFT) reports its transfer stage to Camel #1



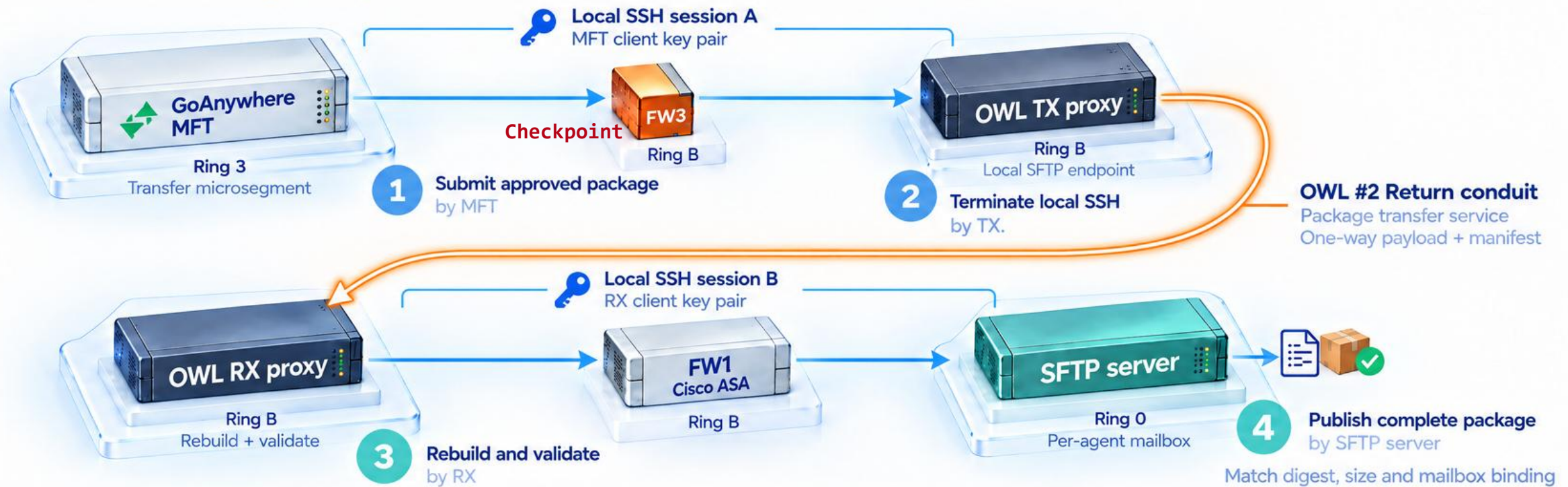
Next: Reports its own stage. Pickup readiness requires confirmed SFTP publication.

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Package landing

Package delivery to the SFTP mailbox

Two firewall crossings. Separate SSH sessions on each side of the OWL data diode proxy.



Staged at MFT & published to SFTP are separate events. Private keys stay local. SSH sessions restart at the receiver.

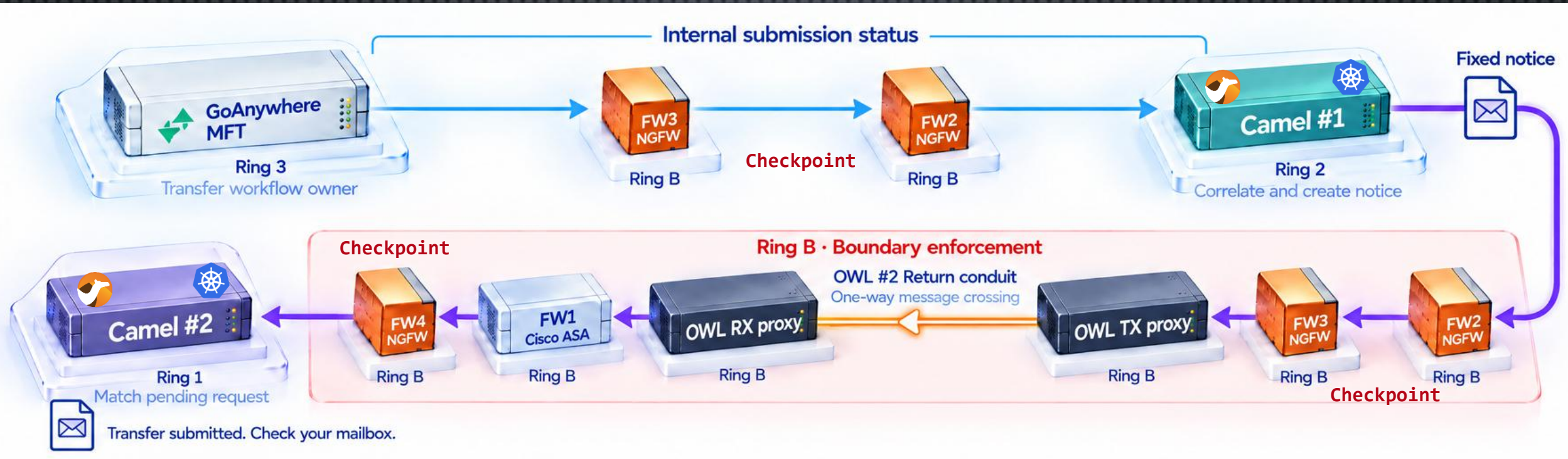
Next: Package delivery to SFTP server

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Package notification

Managed File Transfer (MFT) submission status & return notice

MFT reports submission. Camel #1 creates the fixed notice for Camel #2



Best effort. Source submission does not confirm SFTP receipt. Due to the data diode. Repeat FW2/FW3 icons show the same controls. Final message delivery to the agent: See slide 06.

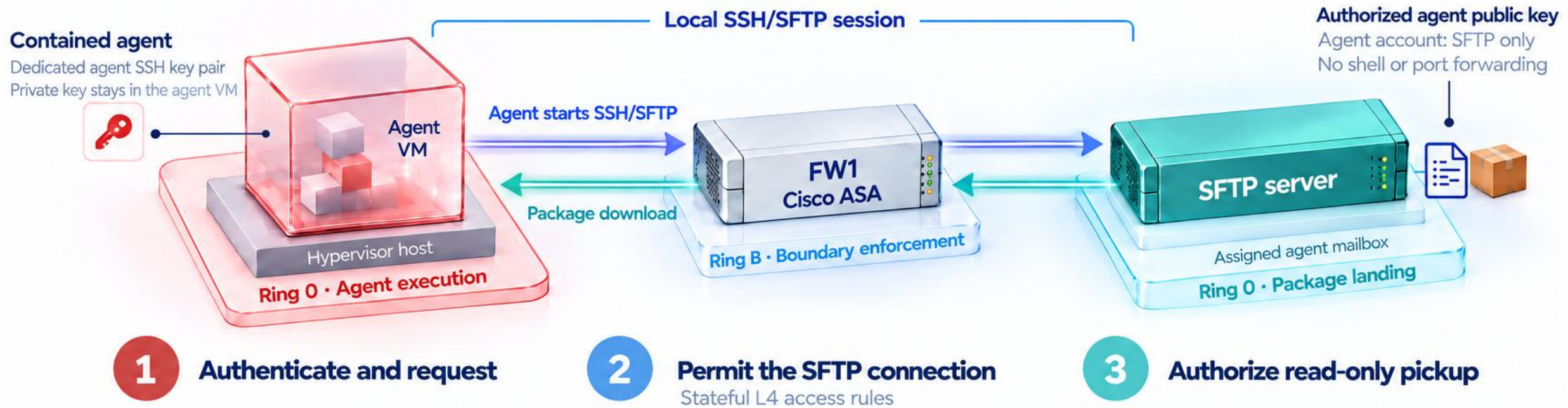
Next: Read only package pickup.

AI CONTAINMENT ASSURANCE LEVELS FOR AGENTS (AI-CAL)

AI-CAL 4.5 | Package pickup

Read only package pickup

The agent authenticates with its SSH key and retrieves its package through Cisco ASA(FW1).



Best effort. Source submission does not confirm SFTP receipt. Due to the data diode. Repeat FW2/FW3 icons show the same controls. Final message delivery to the agent: See slide 06.

Agent permissions: read only in its own mailbox. No Upload, overwrite, delete or rename. Receiver deposit access uses a separate Identity.